ECGR 5411 Assignment 9

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Problem 1

Consider the transfer function

$$F(s) = \frac{K}{s^2 + as + 2}$$

Analytically compute the resonant peak (M_r), resonant frequency (ω_r), and bandwidth (BW) of the system for the following cases:

(a) $a = 2$, $K = 2$

(b) $a = 2$, $K = 10$

(c) $a = 3$, $K = 10$

A tabular view of results is included after the code blocks (Table 1).

```
clc;clear;close all; syms s;
a=[2,2,3]; K=[2,10,10];
wn = sqrt(2);
zeta = (a/(2*wn))'
```

```
zeta = 3x1
    0.7071
    0.7071
    1.0607
```

```
Fs = K ./ (s.^2+a*s+2);

Mr = peakRes(K,wn,zeta)'
```

`peakRes()` *is a user-defined function shown in Listing 1.*

```
Mr = 3x1
    1.0000
    5.0000
    5.0000
```

```
wr = resFreq(zeta,wn)
```

```
wr = 3x1
     0
     0
     0
```

`resFreq()` *is also shown in Listing 1.*

```
wBW = bwFreq(zeta,wn)
```

```
wBW = 3x1
    1.4142
    1.4142
    0.8376
```

Finally, `bwFreq()` *is also included in Listing 1.*

Table 1: Tabular View of Results, Problem 1

Part	M_r	ω_r	BW
(a)	1	0	$\sqrt{2}$
(b)	5	0	$\sqrt{2}$
(c)	5	0	0.8376

Problem 2

Generate the Bode plots using Matlab for Problem 1 (for all three parts). Then manually find the resonant peak (M_r), resonant frequency (ω_r), and bandwidth (BW) from these generated plots. Verify that your answer matches the answers you found analytically in Problem 1.

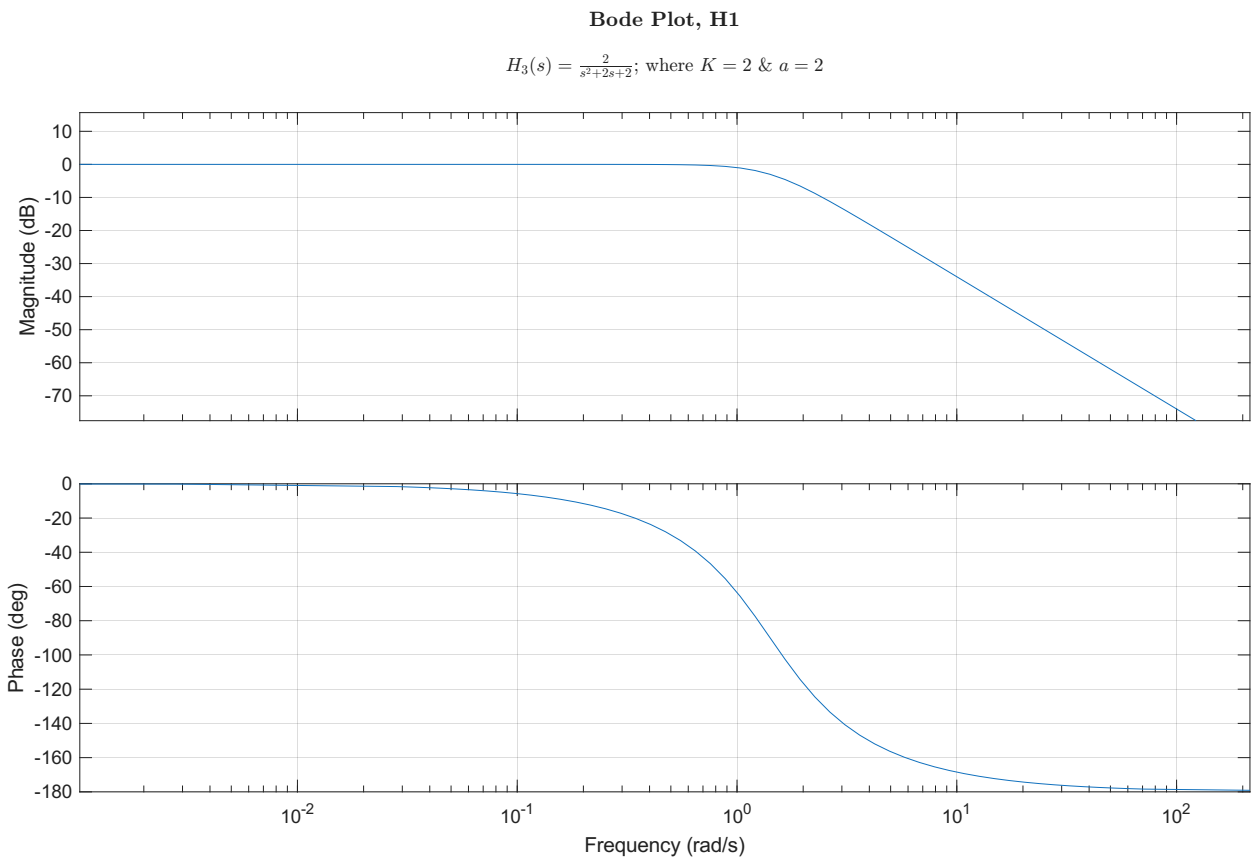
```
% code is continued from above
H1 = tf([0 0 K(1)],[1 a(1) 2])
```

```
H1 =

        2
  -------------
  s^2 + 2 s + 2

Continuous-time transfer function.
Model Properties
```

```
bode(H1);grid on;title('\textbf{Bode Plot, H1}','Interpreter','latex');
subtitle('$H_3(s) = \frac{2}{s^2+2s+2}$; where $K=2$ \& $a=2$','Interpreter','latex');
```



```
% manually computing Mr, wr, and BW
[mag1,phase1,wOut1] = bode(H1,logspace(-10, 2, 2000));

[Mr_dB1, peakIndex1] = max(20*log10(mag1));
Mr1 = mag1(peakIndex1) % in linear units, not dB
```

```
Mr1 = 1.0000
```

```
w_r1     = wOut1(peakIndex1)
```

```
w_r1 = 2.2604e-10
```

```
% low frequency gain using first point
G0_dB1 = 20*log10(mag1(1));

target_dB1 = G0_dB1 - 3;

% Find first frequency where magnitude falls below target_dB
bwIndex1 = find(20*log10(mag1) <= target_dB1, 1, 'first');

w_bw1 = wOut1(bwIndex1)
```

```
w_bw1 = 1.4160
```

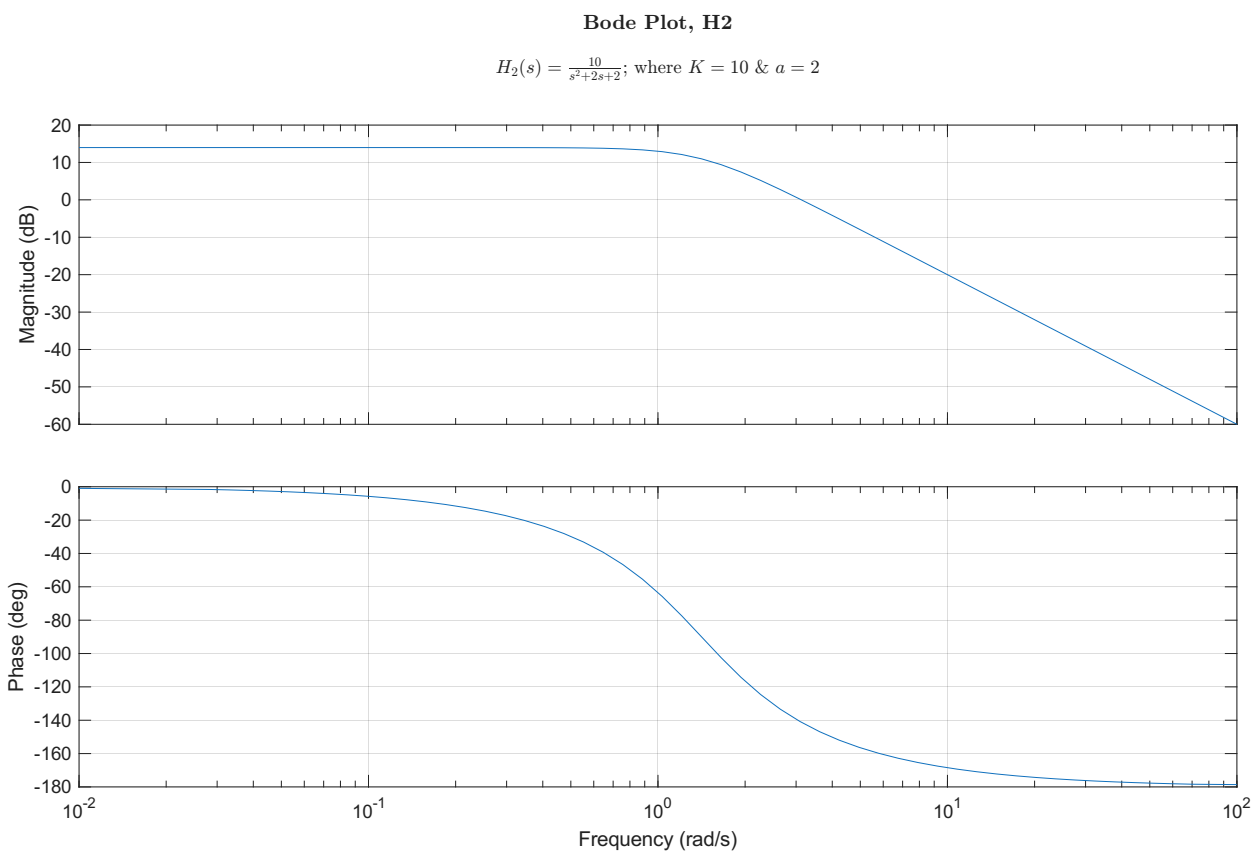
```
H2 = tf([0 0 K(2)],[1 a(2) 2])
```

```
H2 =

       10
  -------------
  s^2 + 2 s + 2

Continuous-time transfer function.
Model Properties
```

```
bode(H2);grid on;title('\textbf{Bode Plot, H2}','Interpreter','latex');
subtitle('$H_2(s) = \frac{10}{s^2+2s+2}$; where $K=10$ \& $a=2$','Interpreter','latex');
```



```
[mag2,phase2,wOut2] = bode(H2,logspace(-10, 2, 2000));

[Mr_dB2, peakIndex2] = max(20*log10(mag2));
Mr2 = mag2(peakIndex2) % in linear units, not dB
```

```
Mr2 = 5.0000
```

```
w_r2 = wOut2(peakIndex2)
```

```
w_r2 = 1.0139e-10
```

```
% low frequency gain using first point
G0_dB2 = 20*log10(mag2(1));

target_dB2 = G0_dB2 - 3;

% Find first frequency where magnitude falls below target_dB
bwIndex2 = find(20*log10(mag2) <= target_dB2, 1, 'first');

w_bw2 = wOut2(bwIndex2)
```

```
w_bw2 = 1.4160
```

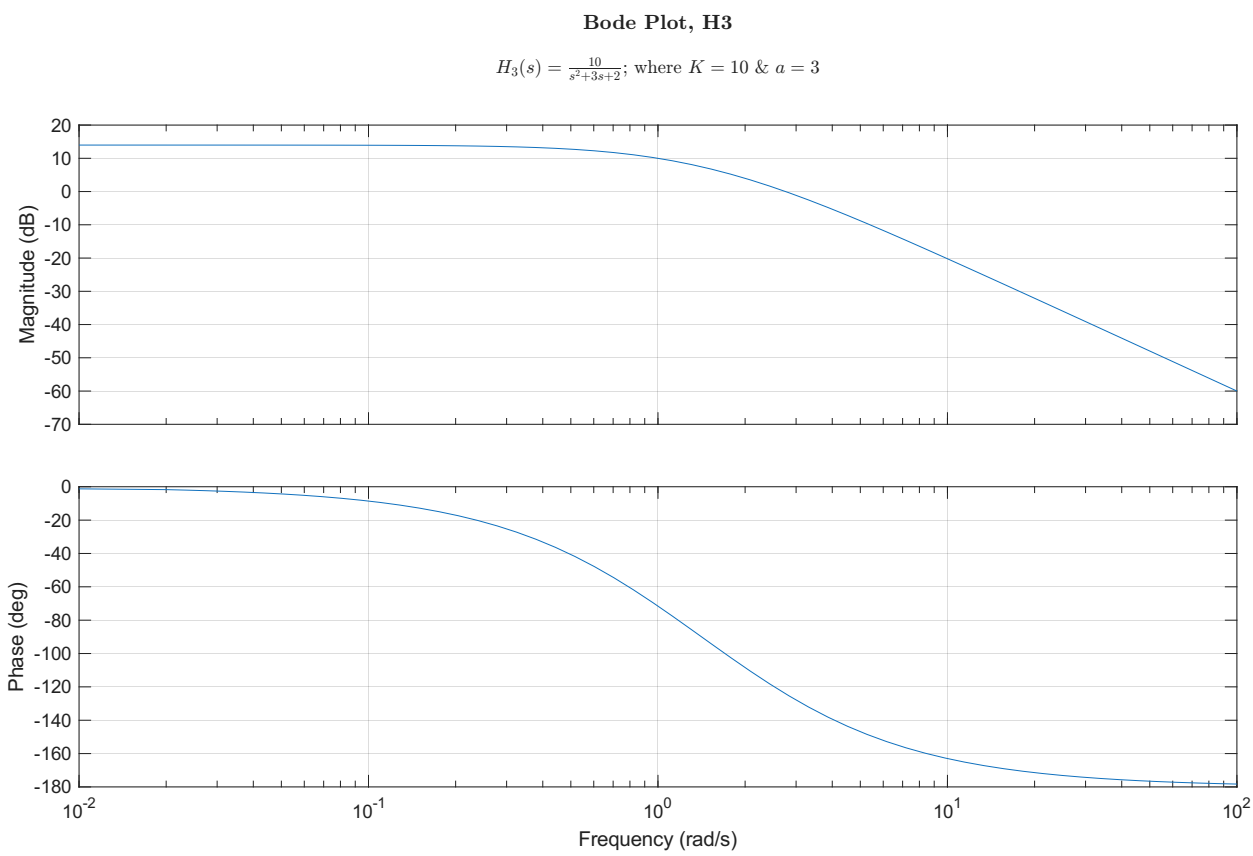
```
H3 = tf([0 0 K(3)],[1 a(3) 2])
```

```
H3 =

       10
  -------------
  s^2 + 3 s + 2

Continuous-time transfer function.
Model Properties
```

```
bode(H3);grid on;title('\textbf{Bode Plot, H3}','Interpreter','latex');
subtitle('$H_3(s) = \frac{10}{s^2+3s+2}$; where $K=10$ \& $a=3$','Interpreter','latex');
```



```
[mag3,phase3,wOut3] = bode(H3,logspace(-10, 2, 2000));

[Mr_dB3, peakIndex3] = max(20*log10(mag3));
Mr3 = mag3(peakIndex3) % in linear units, not dB
```

```
Mr3 = 5
```

```
w_r3 = wOut3(peakIndex3)
```

```
w_r3 = 1.0000e-10
```

```
% low frequency gain using first point
G0_dB3 = 20*log10(mag3(1));

target_dB3 = G0_dB3 - 3;

% Find first frequency where magnitude falls below target_dB
bwIndex3 = find(20*log10(mag3) <= target_dB3, 1, 'first');

w_bw3 = wOut3(bwIndex3)
```

```
w_bw3 = 0.8375
```

Table 2: Tabular View of Results, Problem 2

Part	M_r	ω_r	BW
(a)	1	≈ 0	$1.4160 \approx \sqrt{2}$
(b)	5	≈ 0	$1.4160 \approx \sqrt{2}$
(c)	5	≈ 0	0.8375

For each Bode plot, the ω_r values are considered to be zero rad/s. These data are sufficiently close to the results in Table 1.

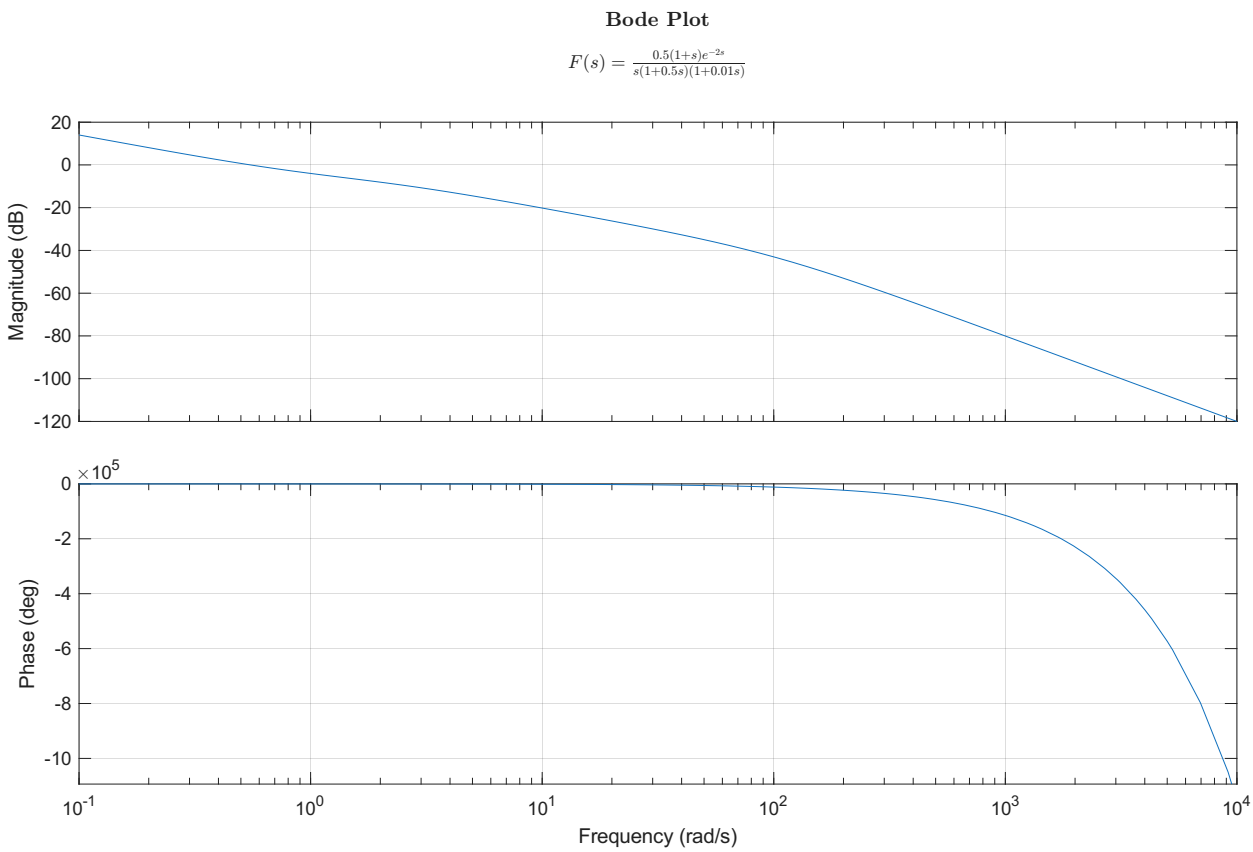
Problem 3

```
clc;clear;close all;
s = tf('s');

% tf without delay
F = 0.5*(1 + s) / (s*(1 + 0.5*s)*(1 + 0.01*s));

% 2-second input delay
F.InputDelay = 2;

% Bode plot
figure;
bode(F);grid on;title('\textbf{Bode Plot}','Interpreter','latex');
subtitle('$F(s) = \frac{0.5(1+s)e^{-2s}}{s(1+0.5s)(1+0.01s)}$','Interpreter','latex');
```




```
[mag,phase,wOut] = bode(F,logspace(-10, 2, 2000));

[Mr_dB, peakIndex] = max(20*log10(mag));
Mr = mag(peakIndex) % in linear units, not dB
```

```
Mr = 5.0000e+09
```

```
w_r = wOut(peakIndex)
```

```
w_r = 1.0000e-10
```

```
% low frequency gain using first point
G0_dB = 20*log10(mag(1));

target_dB = G0_dB - 3;
```

```
% Find first frequency where magnitude falls below target_dB
bwIndex = find(20*log10(mag) <= target_dB, 1, 'first');

w_bw = wOut(bwIndex)
```

```
w_bw = 1.4128e-10
```

As the Bode plot starting frequency decreased (the first input to `logspace(-10,2,2000)`), M_r became sufficiently large, and both ω_r and BW became sufficiently small. This makes sense because of the transfer function's pole at the origin.

$$\therefore \boxed{M_r \to \infty\,,\ \omega_r = 0\,,\ \text{and}\ \ BW = 0.}$$

Function 1: User-Defined Functions for Resonance Parameters

```
% peak resonance
function Mt = peakRes(A,wn,z)
    if z >= 1/sqrt(2)
        Mt = abs(A)/wn^2;
    else
        Mt = abs(A) ./ (wn.^2*2.*z.*sqrt(1-z.^2));
    end
end

% resonant frequency
function wr = resFreq(z,wn)
    if z >= 1/sqrt(2)
        wr = zeros(size(z));
    else
        wr = wn*sqrt(1-2*z.^2)';
    end
end

% bandwidth frequency
function wBW = bwFreq(z,wn)
    wBW = wn.*((1-2*z.^2)+sqrt((2*z.^2-1).^2+1)).^(1/2);
end
```